

PHASE 4

Project Planning Phase

EduGenie - AI Powered Learning Assistant

Plan implementation tasks, dependencies, milestones, responsibilities, testing gates, and deployment readiness.

Prepared as a phase-wise academic submission document based on the supplied EduGenie-Ai project archive.

1. Work Breakdown Structure

1	Environment setup	Virtual environment + dependencies	Python installed
2	Backend foundation	FastAPI app + routing	Requirements installed
3	Data models	Pydantic request/response models	Backend foundation
4	AI integration	Gemini client/service helper	API key configured
5	Learning modules	QA, explanation, quiz, summary, learning path modules	Data models + AI integration
6	Frontend template	index.html controls and result containers	Route contract known
7	Frontend behavior	JavaScript task switching, fetch, quiz interaction	HTML IDs + routes
8	Styling	Responsive CSS	HTML structure
9	Testing	Unit/API/UI verification	All modules integrated
10	Documentation & demo	Final report + screenshots/demo script	Stable build

2. Suggested Milestone Plan

M1	Foundation	App starts and /health responds
M2	Core AI services	Each task module can be invoked independently
M3	UI integration	Five tasks can be triggered from browser
M4	Quality gate	Validation, errors, and quiz rendering work
M5	Submission readiness	Documents, screenshots, test evidence, and demo flow prepared

3. Resource Plan

- Software: Python, VS Code, browser, terminal, pip/venv.
- Libraries: FastAPI, Uvicorn, Jinja2, python-dotenv, google-genai, Pydantic, httpx, pytest.
- External dependency: Google Gemini API access.

4. Risk Register

Missing source modules	Application cannot start	Restore/implement every module imported by main.py before final test
Empty HTML template	UI cannot render expected controls	Complete index.html to match JavaScript element IDs
Invalid/expired API key	AI features fail	Set a valid environment variable and test a minimal API call
Unexpected AI schema	Quiz/path rendering errors	Validate model output and add fallback handling
Network/API limits	Intermittent requests fail	Surface clear errors and avoid excessive repeated calls
Secrets exposed in archive	Security issue	Rotate exposed credentials and remove .env from public submission packages

5. Definition of Done

- All imported modules exist and pass import checks.
- Frontend element IDs match JavaScript selectors.
- All five AI tasks work end-to-end with a valid configuration.
- Automated tests cover health endpoint and core request behavior.
- Manual browser testing is recorded with screenshots.
- Submission documents are ordered and stored phase-wise.

6. Planning Conclusion

The planning phase turns the architecture into a controlled sequence of work with explicit dependencies and quality gates. This reduces the risk of integrating an incomplete frontend or backend late in the project lifecycle.